

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8									
A																							
B																							
C																							
D																							
E																							
F																							
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING H01																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale		NTS		= H01	
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING		Customer		PT Multimas Nabati Asahan		& 304			
00		Issued for Approval		15.01.19		YR										Checked		AW		Drawing Nbr.		18.042-MNA-304.08	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				Customer Doc. Nbr.				Cont.		28	
1		2		3		4		5		6		7		8									

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

	1	2	3	4	5	6	7	8						
A	SHEET INDEX								A					
	PAGE			PAGE NAME				REVISION						
	==MVD-27=H01&304/1			COVER SHEET				00						
	==MVD-27=H01&304/2			Index of Sheet				00						
	==MVD-27=H01&304/3			REFERENCE DESIGNATIONS				00						
	==MVD-27=H01&304/4			REFERENCE DESIGNATIONS				00						
	==MVD-27=H01&304/5			REFERENCE DESIGNATIONS				00						
	==MVD-27=H01&304/6			REFERENCE DESIGNATIONS				00						
	==MVD-27=H01&304/7			DISTRIBUTION OF AUXILIARY CIRCUITS				00						
	==MVD-27=H01&304/8			AUXILIARY CIRCUIT				00						
		==MVD-27=H01&304/9			CIRCUIT BREAKER CONTROL				00					
==MVD-27=H01&304/10			AUXILIARY MOTOR & RELAY				00							
==MVD-27=H01&304/11			DIGITAL INPUT/OUPUT RELAY				00							
==MVD-27=H01&304/12			SPARE CONTACT				00							
==MVD-27=H01&304/13			SPARE CONTACT				00							
==MVD-27=H01&304/14			CURRENT TRANSFOMER				00							
		==MVD-27=H01&304/15			VOLTAGE TRANSFORMER				00					
	==MVD-27=H01&304/16			LV DOOR LAYOUT				00						
	==MVD-27=H01&304/17			LV BOX LAYOUT				00						
	==MVD-27=H01&304/18			PART LIST				00						
	==MVD-27=H01&304/19			Terminal diagram ==MVD-27=H01+-XDA										
	==MVD-27=H01&304/20			Terminal diagram ==MVD-27=H01+-XDB1										
	==MVD-27=H01&304/21			Terminal diagram ==MVD-27=H01+-XDC1										
	==MVD-27=H01&304/22			Terminal diagram ==MVD-27=H01+-XDC2										
	==MVD-27=H01&304/23			Terminal diagram ==MVD-27=H01+-XDI										
	==MVD-27=H01&304/24			Terminal diagram ==MVD-27=H01+-XDI2										
	==MVD-27=H01&304/25			Terminal diagram ==MVD-27=H01+-XDI3										
	==MVD-27=H01&304/26			Terminal diagram ==MVD-27=H01+-XDI6										
	==MVD-27=H01&304/27			Terminal diagram ==MVD-27=H01+-XDV										
	==MVD-27=H01&304/28			Terminal diagram ==MVD-27=H01+-XDX										
E									E					
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant		Customer				PT Multimas Nabati Asahan	& 304	
00	Issued for Approval	15.01.19	YR	Checked	AW			Drawing Nbr.	18.042-MNA-304.08	Sh.		2		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.		Cont.		28		
	1	2	3	4	5	6	7	8						

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8							
A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A					
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR												
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR												
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.												
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT												
B	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							B					
		-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT											
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT												
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT												
C	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB3	HEATER LOCATED IN MOTOR CABINET							C					
	-BGK	POSITION SWITCH OF KEY LOCK						-EB4	HEATER FOR VENTILATION TEST												
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.												
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE												
D	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D					
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1												
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2												
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3												
E	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							E					
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1												
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2												
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3												
F	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							F					
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS												
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1												
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2												
G	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							G					
	-BM	HYGROSTAT						-FCT	THERMAL OVERLOAD RELEASES												
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS												
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR												
H	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							H					
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK												
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE												
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE												
I	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							I					
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFC	CLOSING RELAYS OR CONTACTORS												
	-BT	THERMOSTAT						-KFI	INTEGRATED CIRCUITS												
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY												
J	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFM	MICROPROCESSORS							J					
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFN	ANTIPUMPING RELAYS												
	-BU	BUCHHOLZ RELAY						-KFO	OPENING RELAYS OR CONTACTORS												
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)												
K	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFR	RECLOSING RELAYS							K					
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFS	SYNCHRONIZING DEVICES												
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS												
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFZ	CONTROLLERS												
L	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							L					
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD												
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1												
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2												
M								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							M					
1		2		3		4		5		6		7		8							
				Date Prep.		15.11.2018		PROJECT TITLE		ABB		EPMV		REFERENCE DESIGNATIONS		Scale		NTS		= H01	
				Prepared		YR										Customer		PT Multimas Nabati Asahan		& 304	
00		Issued for Approval		15.01.19		YR		Checked		AW		PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.		18.042-MNA-304.08		Sh. 4	
Ind.		Revision		Date		Name		Approved		AI						Based on:		BE318042		Customer Doc. Nbr.	

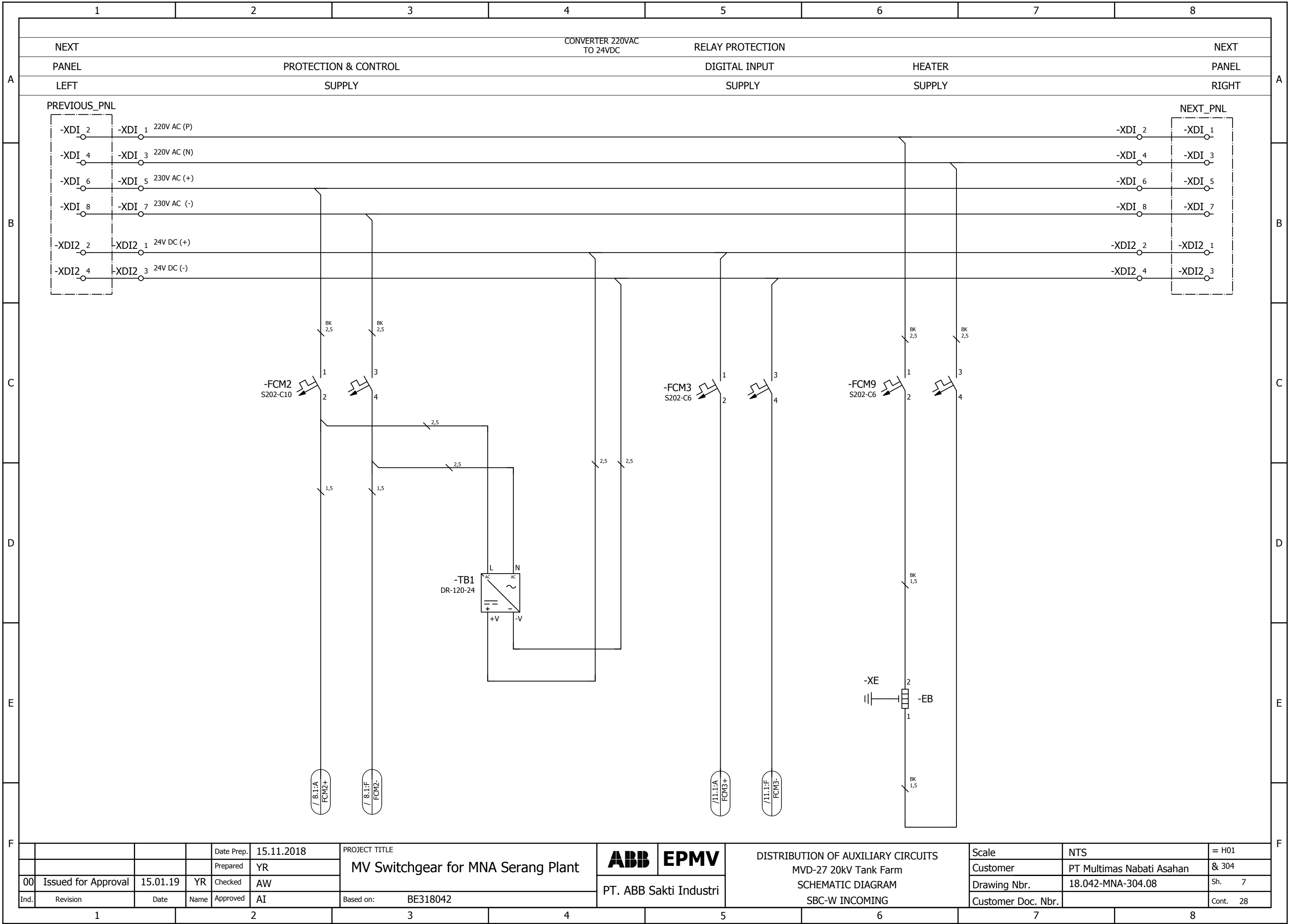
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8				
A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							A		
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS									
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS									
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS									
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK									
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING									
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH									
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES									
	B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							B	
		-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES								
-PFB		BLUE SIGNAL LAMPS						-RAI	INDUCTORS									
-PFF		FLAG RELAYS						-RAR	RESISTORS									
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS									
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE									
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION									
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD									
	C	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							C	
		-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2								
-PFY		YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2									
-PGA		AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS									
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4									
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5									
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES									
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS									
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS									
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS									
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS									
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES									
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS									
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS									
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES									
	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS									
	-PGV	VOLTMETERS						-TB	CONVERTER									
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE									
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE									
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE									
E	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							E		
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE									
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER									
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)																
	-QBD4	TWO-WAY DISCONNECTOR																
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER																
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-27 20kV Tank Farm		Scale	NTS		= H01	F	
					Prepared	YR							Customer	PT Multimas Nabati Asahan		& 304		
	00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W INCOMING		Drawing Nbr.	18.042-MNA-304.08		Sh. 5		
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042					Customer Doc. Nbr.			Cont. 28		
1		2		3		4		5		6		7		8				

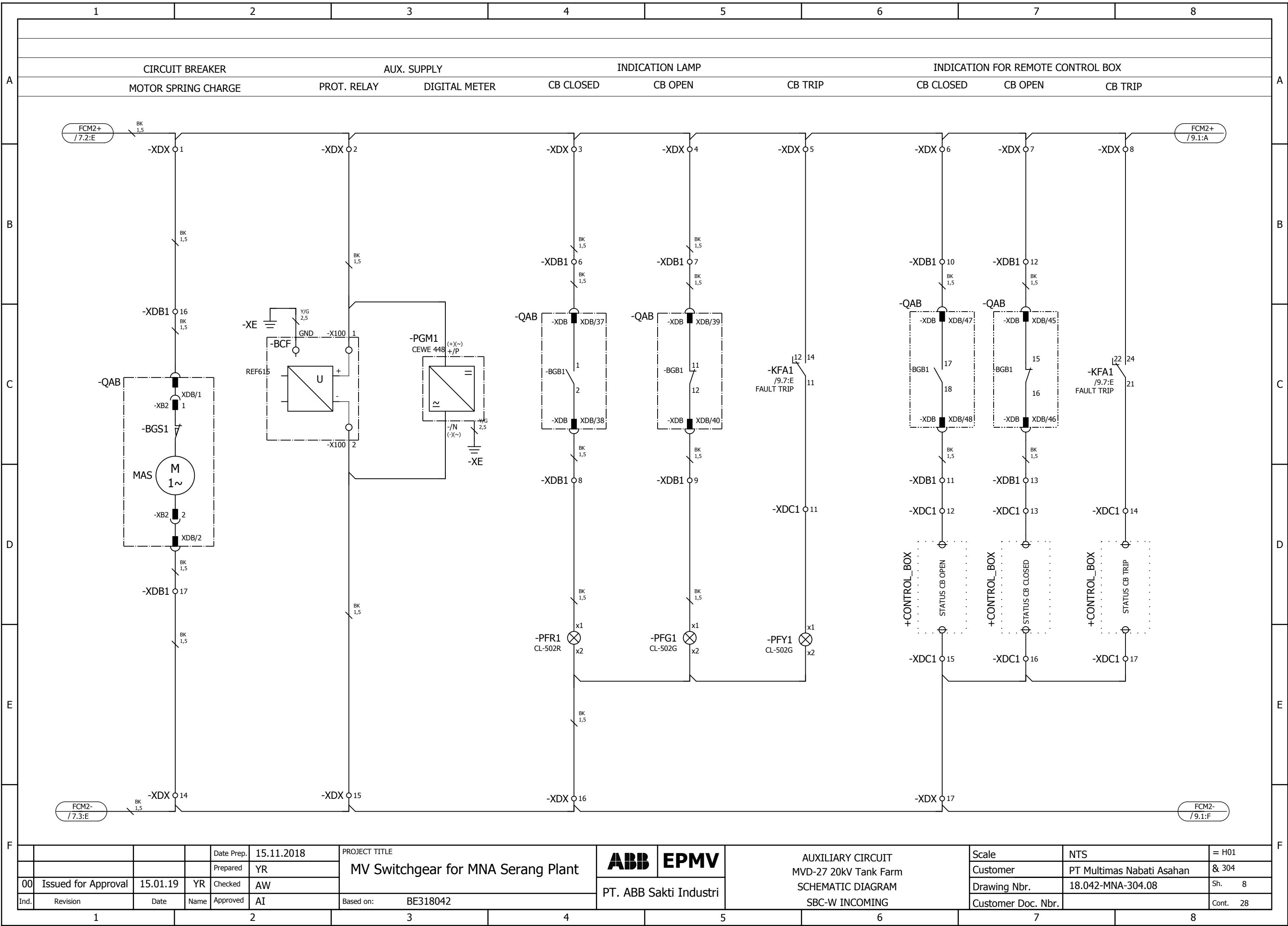
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

A	-TFA	ACTIVE POWER TRANSDUCERS	A		
	-TFC	CURRENT TRANSDUCERS			
	-TFF	FREQUENCY TRANSDUCERS		A	
	-TFJ	ACTIVE ENERGY TRANSDUCERS			
	-TFK	REACTIVE ENERGY TRANSDUCERS			
	-TFM	MULTIFUNCTION TRANSDUCERS			
	-TFP	POWER-FACTOR TRANSDUCERS			
	-TFQ	REACTIVE POWER TRANSDUCERS			
	-TFS	SIGNAL CONVERTERS			
	-TFV	VOLTAGE TRANSDUCERS			
	-WA	BUSBARS			B
	-WBC	POWER CABLES			
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.			
	-WGC	CONTROL CABLES			
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)			
	-WH	OPTICAL FIBRES			
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS			
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2			
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			
-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				
-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS				
B	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	B		
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS			
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS			
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR			
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			
	-XDC	CUSTOMER TERMINAL BLOCK			
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR			
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS			
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			
-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS				
-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS				
-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)				
-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)				
C	-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	C		
	-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING			
	-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION			
	-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS			
	-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS			
	-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS			
	-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS			
	-XDI9	TERMINAL BLOCK FOR SPECIAL USE			
	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD			
	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS			
	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT			
	-XDS	CURRENT SOCKETS			
	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB			
	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK			
	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK			
	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS			
	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1			
	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2			
	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3			
	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR			
-XDX	SUPPORT TERMINAL BLOCK				
-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE				
-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE				
-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE				
-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE				
-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE				
-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE				
-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT				
D			D		
	E				E
F				F	

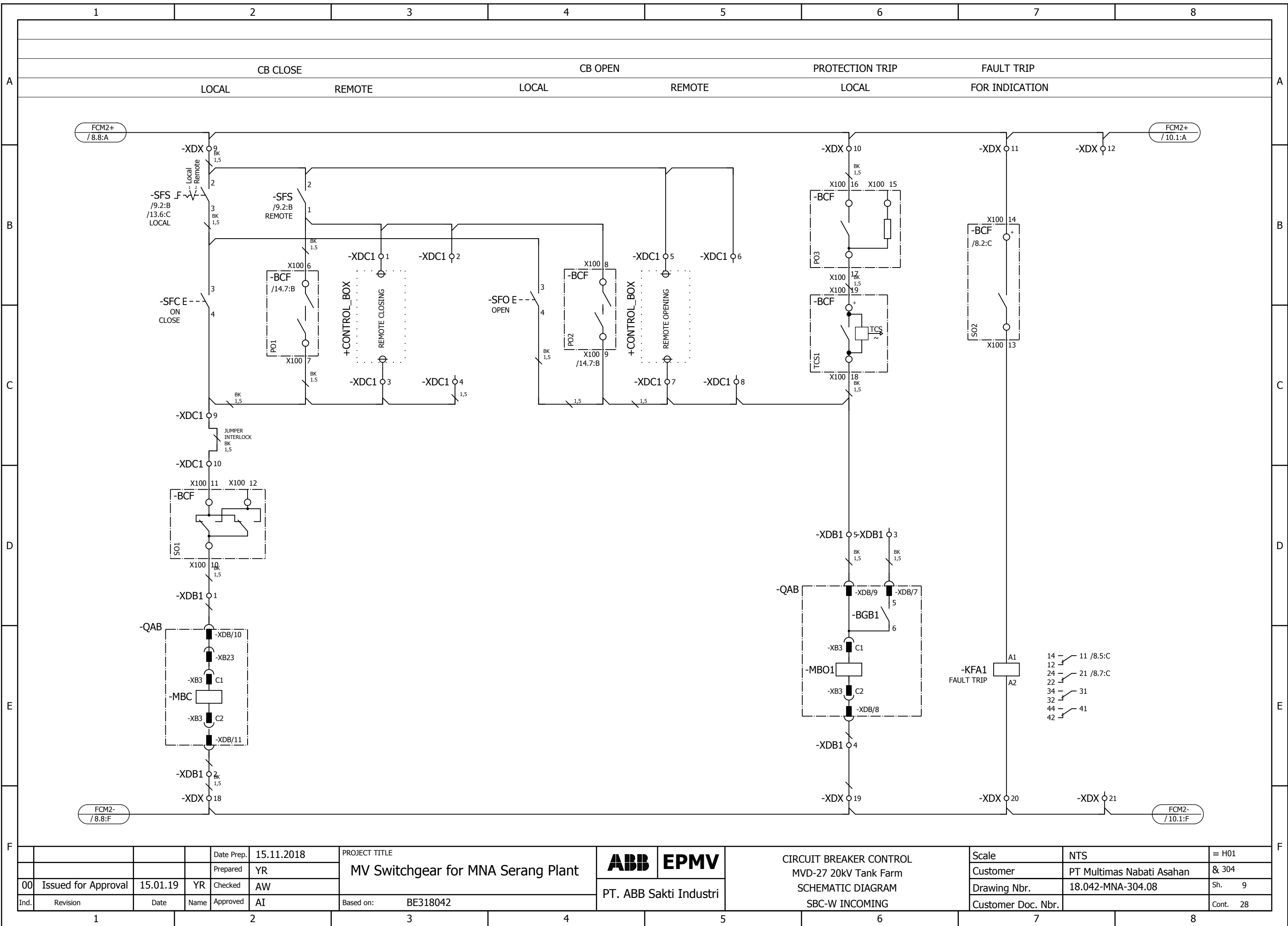
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

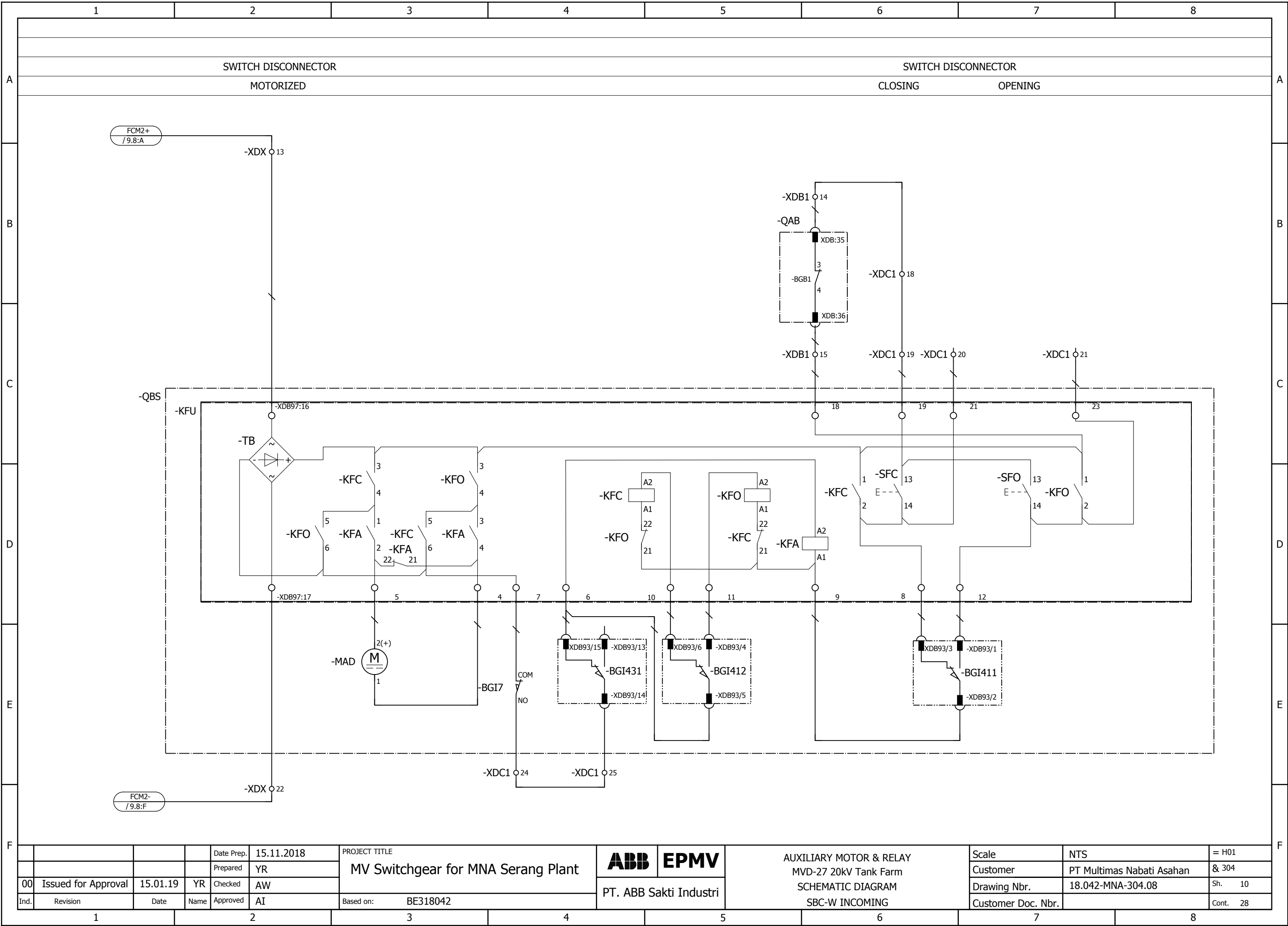


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

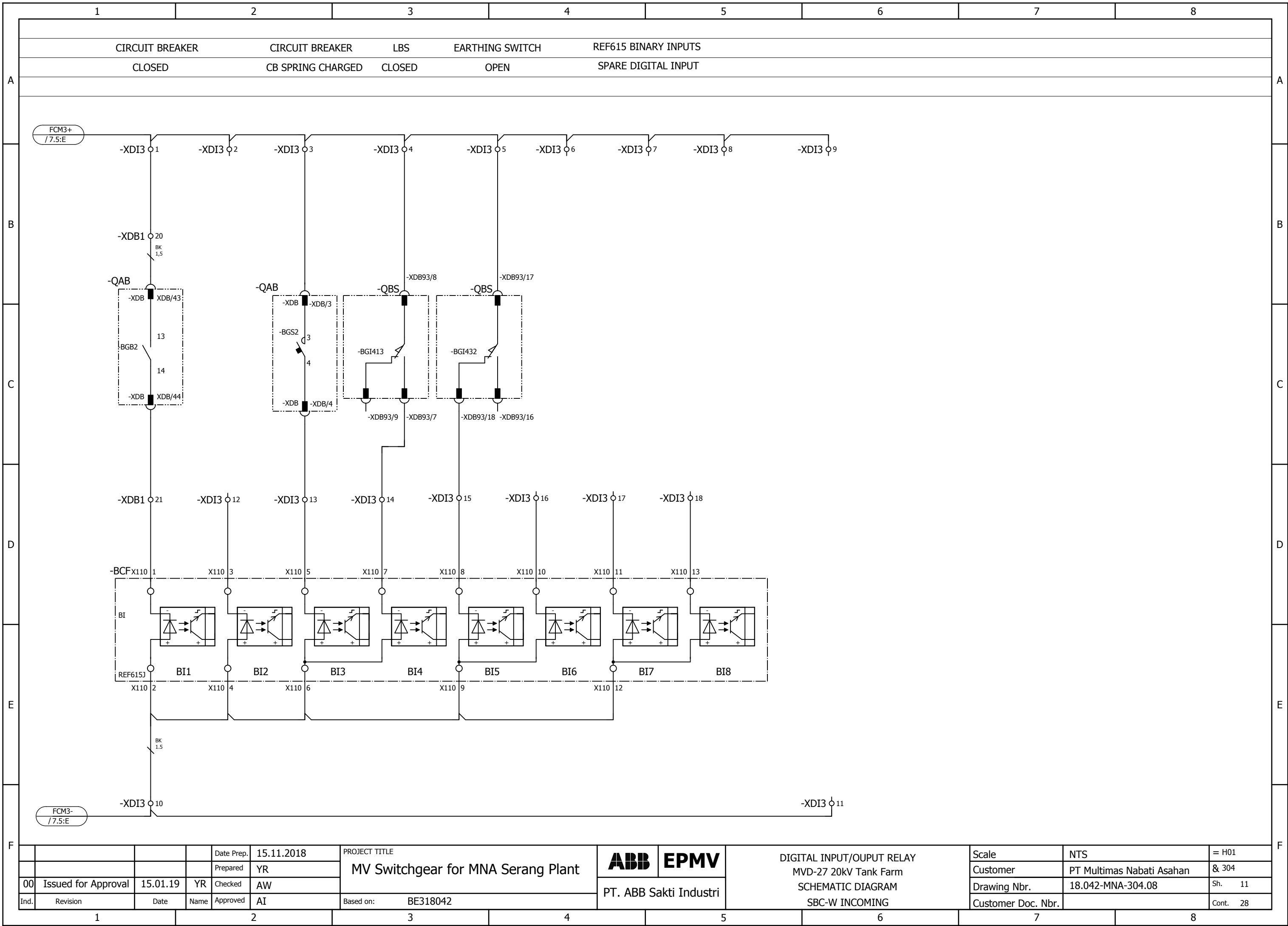


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	CIRCUIT BREAKER CONTROL MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-304.08		Sh. 9		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

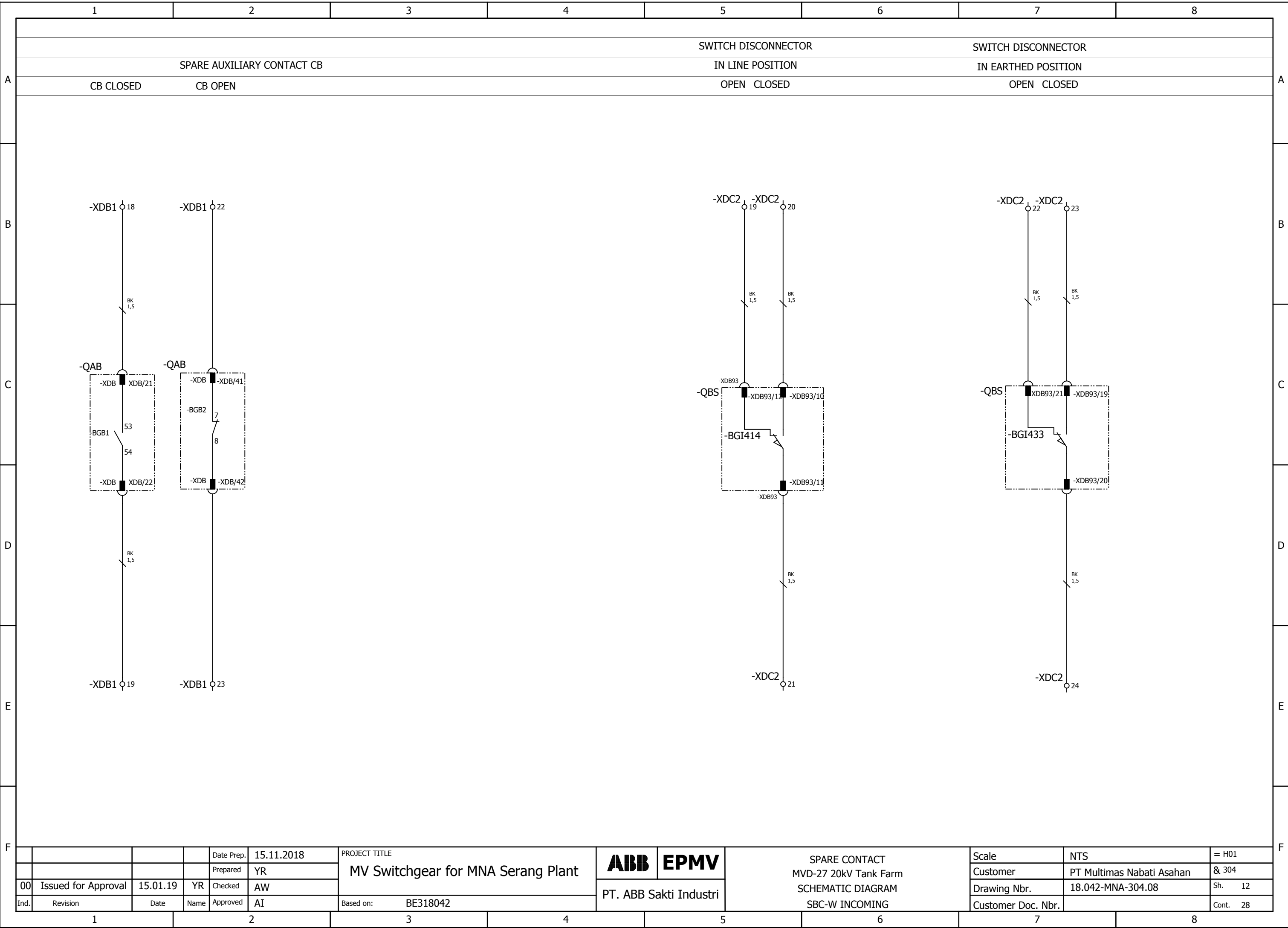


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



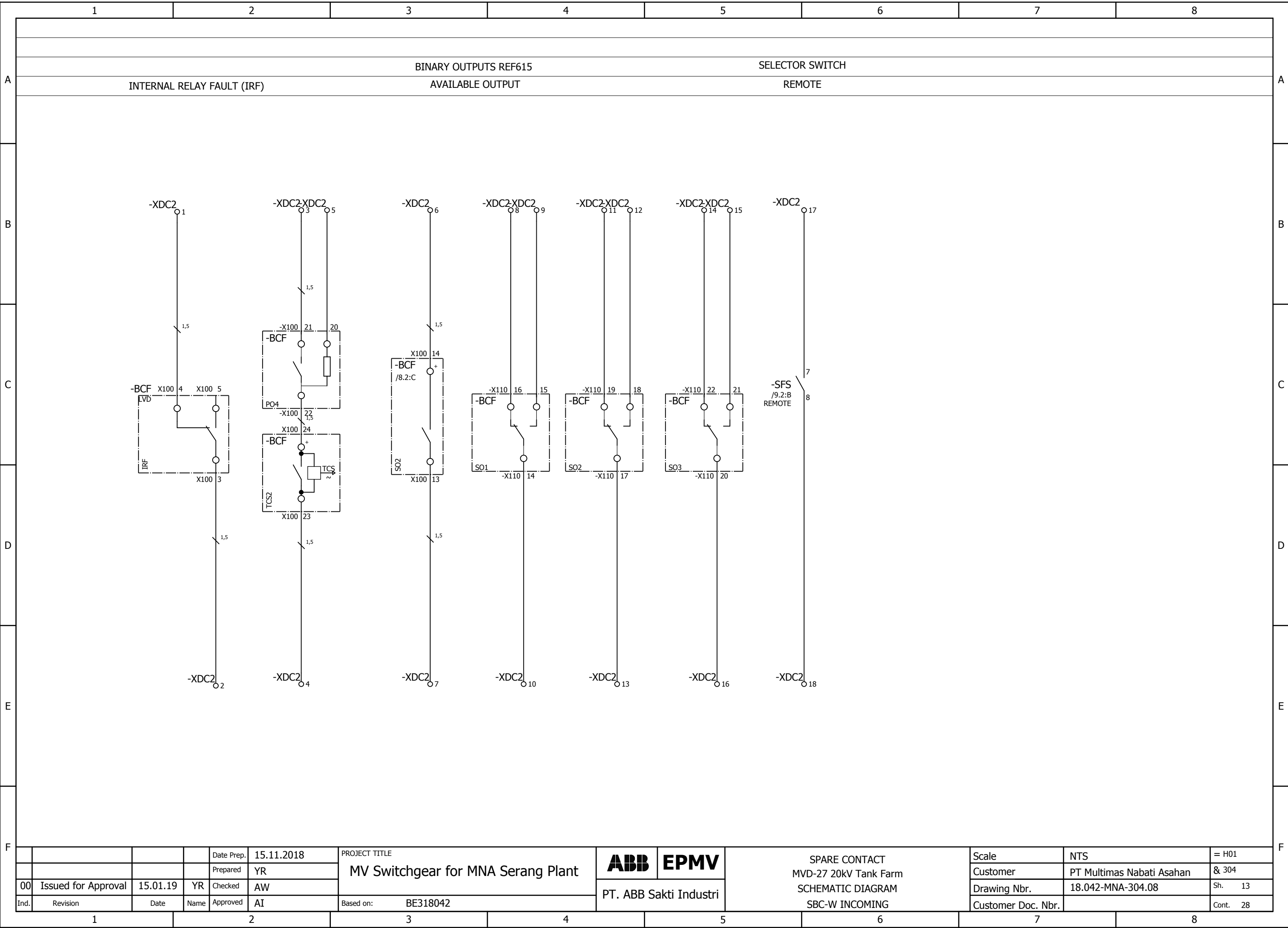
				Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING		Scale	NTS	= H01
				Prepared	YR						Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW	Based on: BE318042		PT. ABB Sakti Industri			Drawing Nbr.	18.042-MNA-304.08	Sh. 11
Ind.	Revision	Date	Name	Approved	AI						Customer Doc. Nbr.		Cont. 28

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

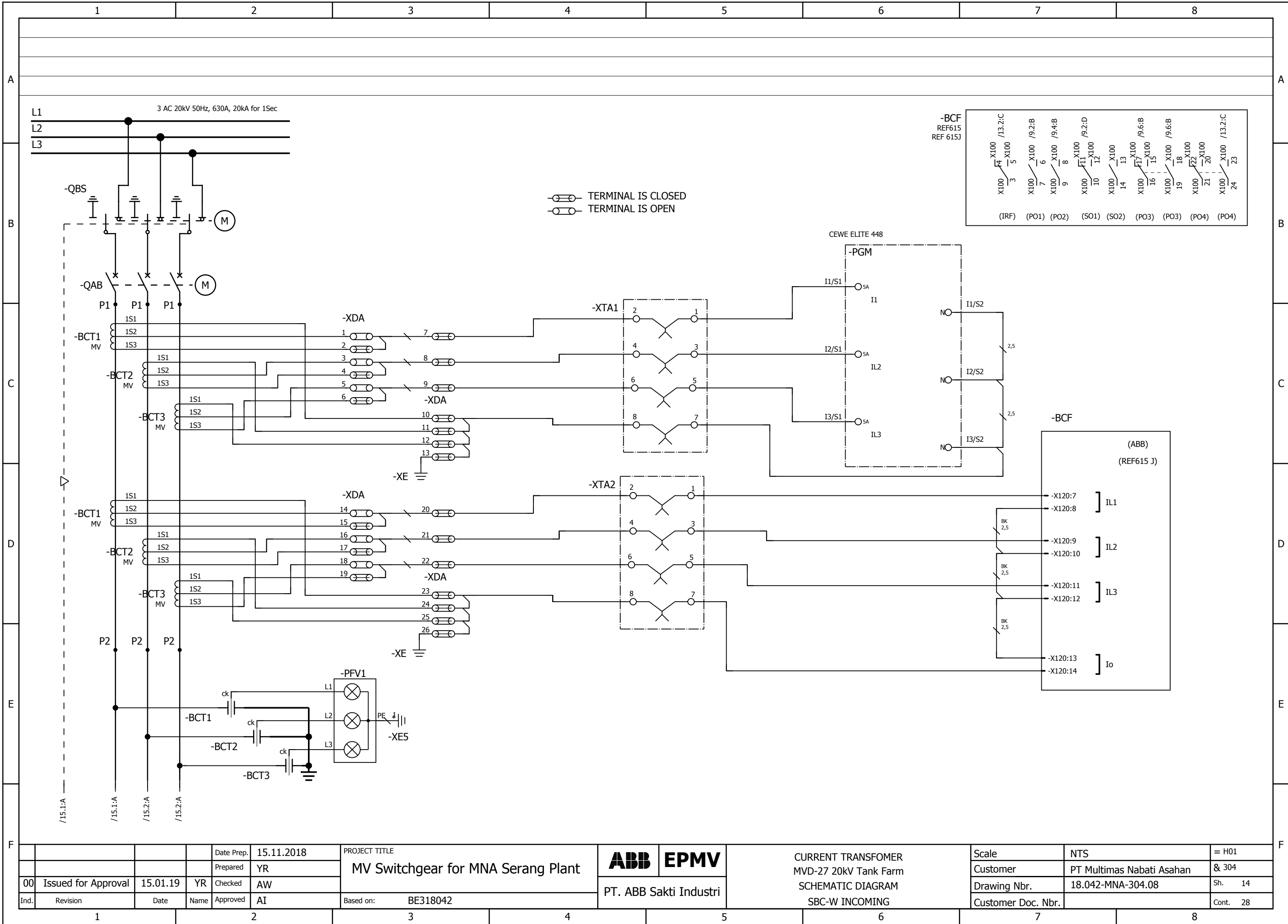


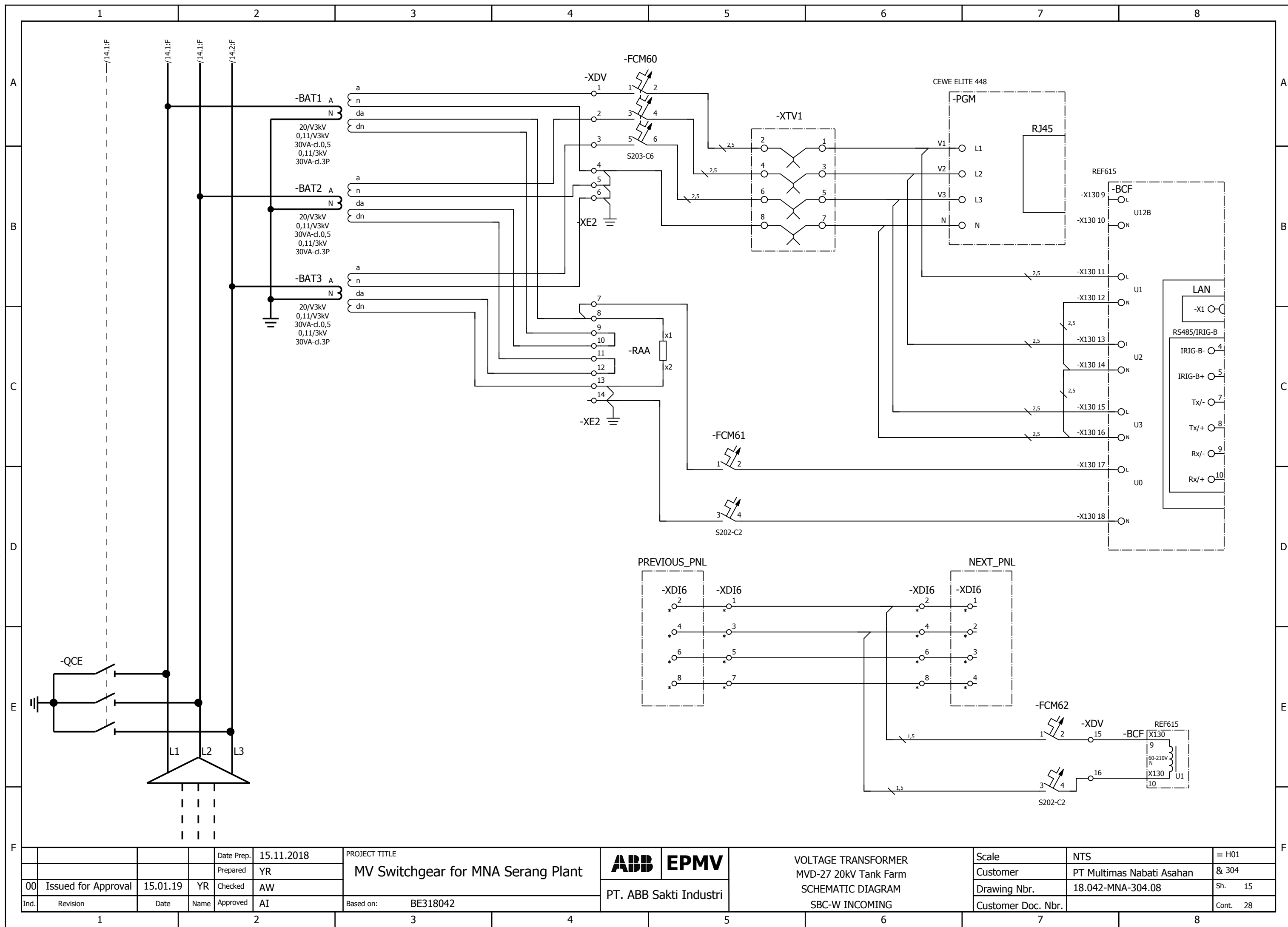
00	Ind.	Revision	Date	Name	Date Prep.	PROJECT TITLE	Based on:	BE318042	ABB	EPMV	PT. ABB Sakti Industri	SPARE CONTACT MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
					15.11.2018								Customer	PT Multimas Nabati Asahan	& 304
					YR								Drawing Nbr.	18.042-MNA-304.08	Sh. 12
					AW								Customer Doc. Nbr.		Cont. 28

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



00	Issued for Approval	15.01.19	YR	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant	Based on:	BE318042	ABB EPMV	PT. ABB Sakti Industri	SPARE CONTACT MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR								Customer	PT Multimas Nabati Asahan	& 304
				Checked	AW								Drawing Nbr.	18.042-MNA-304.08	Sh. 13
				Approved	AI								Customer Doc. Nbr.		Cont. 28





No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	PART LIST MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-304.08	Sh. 18	
Ind.	Revision	Date	Name	Approved	AI			Based on: BE318042		Customer Doc. Nbr.		Cont. 28

[illegible]

F														F	E	D	C	B	A		

F										E		D		C		B		A			
				Date Prep.	15.11.2018	PROJECT TITLE				PT. ABB Sakti Industri	Terminal diagram ==MVD-27=H01+-XDC1				Scale			= H01			
				Prepared	YR	MV Switchgear for MNA Serang Plant									Customer		PT Multimas Nabati Asahan		& 304		
				Checked	AW										Drawing Nbr.				Sh. 21		
Ind.	Revision		Date	Name	Approved	AI	Based on: BE318042								Customer Doc. Nbr.				Cont. 28		
1		2			3			4			5			6			7			8	

==MVD-27=H01+-XDC1																			
REMARK	CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK										
BK 1.5 mm ²		+CONTROL_BOX		1 ●	PT 2.5		-SFS:1												
				2 ●	PT 2.5		-BCF:X100:8												
		-BCF:X100:7		3 ●	PT 2.5		+CONTROL_BOX												
				4 ●	PT 2.5														
1,5 mm ²		+CONTROL_BOX		5 ●	PT 2.5		-SFS:2												
				6 ●	PT 2.5		+CONTROL_BOX												
		-BCF:X100:9		7 ●	PT 2.5														
		-BCF:X100:18		8 ●	PT 2.5														
BK 1,5 mm ²				9 ●	PT 2.5		-SFC:4												
		-BCF:X100:11		10 ●	PT 2.5														
		-KFA1:11		11	PT 2.5		-PFY1:X1												
		-XDB1:11		12	PT 2.5		+CONTROL_BOX												
BK 1,5 mm ²		-XDB1:13		13	PT 2.5		+CONTROL_BOX												
		-KFA1:21		14	PT 2.5		+CONTROL_BOX												
		+CONTROL_BOX		15 ●	PT 2.5		-XDX:17												
		+CONTROL_BOX		16 ●	PT 2.5														
BK 1,5 mm ²		+CONTROL_BOX		17 ●	PT 2.5														
				18 ●	PT 2.5		-XDB1:14												
		-QBS-KFU-XDB97:19		19 ●	PT 2.5														
		-QBS-KFU-XDB97:21		20	PT 2.5														
BK 1,5 mm ²		-QBS-KFU-XDB97:23		21	PT 2.5														
				24 ●	PT 2.5		-QBS-BGI7:NO												
				25 ●	PT 2.5		-QBS:-XDB93/14												
								</											

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

F										E										D										C										B										A									